ANNEX B

Cover page for INNOVA JUNIOR COLLEGE to submit with Prelim Exam Paper H2 Maths (9740)

Qn/No	Topic Set	Paper No.	Answers	Marks
1	System of Equations	9740/1	20.8, 36.8, 48.8	4
2	Integration	9740/1	(i) $\left(\frac{1}{2}\cos\frac{x}{2}\right)e^{\sin\frac{x}{2}}$ (ii) $\left(4\sin\frac{x}{2}\right)e^{\sin\frac{x}{2}}-4e^{\sin\frac{x}{2}}+C$	5
3	Complex Numbers	9740/1	(ii) $\alpha=\frac{\pi}{3}$	5
4	Transformation of Graphs	9740/1		6
5	Induction	9740/1		7
6	Differential Equation	9740/1	$y=\frac{e^{\pi x}}{\sqrt{3}}\sin\left(\sqrt{3}x+\frac{\pi}{3}\right)$	7
7	Binomial Series	9740/1	$1-x+x^2+.....$ $\lvert x\rvert<\frac{1}{2}$ or $-\frac{1}{2}<x<\frac{1}{2}$ $\frac{31}{18}$	8
8	Sequences and Series	9740/1	$k=\frac{1}{6}$ 17 days	8
9	Maclaurin's Series	9740/1	$-3x-\frac{9}{2}x^2-\frac{9}{2}x^3$ $1-\frac{9}{2}x^2$	8
10	Functions	9740/1	(a) $R_f=(-1.23,1.23)$ ff does not exist $x=y-2\tan^{-1}(2y)$ (b) $g(x)=(x-3)^2+4$	9
11	Application of Differentiation	9740/1	$\frac{dy}{dx}=\frac{n(3y-5nx)}{(5y-3nx)}$ $\left(\frac{5}{2n},\frac{3}{2}\right)$ and $\left(-\frac{5}{2n},-\frac{3}{2}\right)$ $y=-\frac{3}{5n}x+\frac{6}{5n^2}$ and $y=-\frac{3}{5n}x-\frac{6}{5n^2}$	10

12	Series (AP, GP)	9740/1	$\alpha = -\ln 4 \ = \ln\frac{1}{4}$ Greatest value of n is 56	10
13	Vectors	9740/1	(i) $\mathbf{r} = 3\mathbf{j} + \lambda(5\mathbf{i} - 3\mathbf{j} + 2\mathbf{k})$ $\mathbf{r} = 5\mathbf{j} + \mu(-5\mathbf{i} + 3\mathbf{j} + 2\mathbf{k})$ (ii) 37.9° (iii) $\mathbf{r} \bullet \begin{pmatrix} 6 \\ 10 \\ -15 \end{pmatrix} = 30$ (iv) $\frac{20}{19}$ units	13

Qn/No	**Topic Set**	**Paper No.**	**Answers**	**Marks**
1	Curve sketching	9740/2	(i) $y = x + 7$, $x = -2$ (ii) $5 - 2\sqrt{2} < k < 5 + 2\sqrt{2}$	9
2	Vectors	9740/2	(a) 12	7
3	Applications of Integration, Transformation	9740/2	(i) $\frac{1}{2}\ln(\frac{7}{4})$ (ii) $\frac{7}{2}\ln(\frac{7}{4}) - 2\ln 2$ (iii) 6.72 unit^3 (iv) $y = \frac{7e^{\frac{1}{3}x}}{e^{\frac{1}{3}x} + 3e^{-\frac{1}{3}x}} - 5$	11
4	Complex Numbers	9740/2	(ii) $z_1 = 2e^{-\frac{\pi}{3}i}$, $z_2 = 2e^{\frac{\pi}{3}i}$ (iii) 2 (iv) 4 (v) 4π	13
Qn/No	**Topic Set**	**Paper No.**	**Answers**	**Marks**
5	Permutation & Combination	9740/2	59	3
6	Poisson Distribution	9740/2	$\mu = -\ln 0.7$	3
7	Sampling Methods	9740/2	No. Because not every individual has an equal chance of being selected. Only the “even numbered” students will have a chance of being selected. Sampling interval $k = 50$. Get a list of all students arranged according to school register. Select the first student at random from the first 50 students. Select every 50$^{\text{th}}$ student	4

			thereafter till a sample of 40 is obtained.	
8	Probability	9740/2	(i) 67/150 (ii) 29/50 (iii) not independent	8
9	Normal Distribution	9740/2	(i) 0.186 (ii) 0.119 (iii) 0.598	9
10	Hypothesis Testing	9740/2	(a) p-value = 0.0412 (to 3 s.f) Since p-value < 0.05, reject H_0 and conclude that there sufficient evidence at 5% significance level, that the inventor has overstated the mean run time. Assume that the run time is normally distributed. (b) $\{\mu_0 \in \mathbb{R} : \mu_0 < 142.82 \text{ or } \mu_0 > 147.18\}$	10
11	Correlation & Regression	9740/2	(i) $r = 0.875$ (iii) a = 3.00 and b = 3.19 (3 s.f.) (iv) $r = 0.961$ $x = 2.6$	10
12	Binomial Distribution	9740/2	(a) $\mathrm{P}(3.8 < \overline{X} < 4.2) = 0.571$ (b) 0.224, 0.875, 0.629	13